CHAPTER 18

CLIMATE OF INDIA

CHAPTER ESSENCE AT A GLANCE:
Master the dynamic mechanisms of the Indian Monsoon, Inter-Tropical Convergence Zone (ITCZ) shift, Subtropical and Tropical Jet Streams, Tibetan plateau heating, ENSO and El Nino impacts, four meteorological seasons, and Western Disturbances.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Thermal and Dynamic Monsoon Genesis:** The Indian monsoon is driven by differential heating of the landmass and Indian Ocean, coupled with the northward migration of the Inter-Tropical Convergence Zone (ITCZ) over the Ganga plain in summer, creating a thermal low drawing cross-equatorial trade winds.

• **Upper Tropospheric Jet Stream Steering:** Withdrawal of the Subtropical Westerly Jet Stream to the north of the Himalayas in June coincides with the sudden burst of the monsoon; establishment of the Tropical Easterly Jet Stream over Peninsular India steers rain-bearing depressions.

• **ENSO Teleconnections and Dipole Dynamics:** El Nino (warming of eastern equatorial Pacific) causes atmospheric pressure to rise over the Indian Ocean, inducing monsoon failure and drought; positive Indian Ocean Dipole (IOD) counteracts El Nino by boosting Indian monsoon rainfall.

• **Four Meteorological Seasons:** Cold weather season (December to February with Mediterranean Western Disturbances), Hot weather season (March to May with Loo and Kalbaisakhi), Southwest monsoon season (June to September), and Retreating monsoon season (October to November).

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Southern Oscillation Index (SOI)} = P_{\text{Tahiti}} - P_{\text{Darwin}} \rightarrow \text{Negative SOI Signals El Nino Drought}$$

Walker Oceanic Atmospheric Teleconnection Barometric Index

$$a = -\frac{1}{\rho}\frac{\Delta P}{\Delta x} \rightarrow \text{Monsoon Pressure Gradient Driving Cross-Equatorial Flow}$$

Hydrodynamic Driving Force of Southwest Monsoon Wind Field

$$\Delta T = \Gamma_d \cdot \Delta z \rightarrow \text{Adiabatic Warming Dries Deccan Leeward Interior (Rain Shadow)}$$

Thermodynamic Dessication of Peninsular Interior

$$\text{Indian Annual Rainfall Budget } \approx 4000 \text{ Billion m}^3 \rightarrow \approx 75\% \text{ in Southwest Monsoon}$$

Seasonal Precipitation Concentration Ratio of Indian Monsoon

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Western Disturbances that bring vital winter rains to north-west India originate over the Mediterranean Sea and are steered into India by the Subtropical Westerly Jet Stream, not the southwest monsoon.

• **Alert:** The Tamil Nadu coast (Coromandel Coast) remains dry during the southwest monsoon because it lies parallel to the Bay of Bengal branch and in the rain shadow of the Arabian Sea branch; it gets rainfall during the retreating northeast monsoon (October to December).

• **Alert:** The 'Burst of Monsoon' refers to the sudden, explosive onset of torrential downpours accompanied by violent thunder and lightning in early June, not ordinary intermittent pre-monsoon showers.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is the 'Burst of Monsoon'? Which Indian state experiences the first burst of the southwest monsoon? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Burst of Monsoon:** The sudden, dramatic onset of heavy, continuous rainfall accompanied by intense thunder and lightning that replaces the dry scorching heat in early June. **[1 Mark]**

• **First State:** The state of Kerala on the southwestern coast is the first mainland state to receive the burst of the southwest monsoon around 1 June. **[1 Mark]**

2. What are Western Disturbances? State their agricultural importance for northern India. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Origin:** Shallow extra-tropical cyclonic depressions that originate over the Mediterranean Sea and travel eastward across West Asia steered by the westerly jet stream. **[1 Mark]**

• **Agricultural Importance:** They cause moderate winter rainfall in Punjab, Haryana, and plains, which is of critical economic value for the cultivation of Rabi crops, especially wheat. **[1 Mark]**

3. Explain the role of the Subtropical Westerly Jet Stream and the Tropical Easterly Jet Stream in the mechanism of the Indian monsoon. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Subtropical Westerly Jet Stream:** In winter, it blows south of the Himalayas across northern India; its sudden northward shift beyond the Tibetan Plateau in late May creates a low-pressure vacuum that permits monsoon onset. **[1.5 Marks]**

• **Tropical Easterly Jet Stream:** Develops in summer at high altitudes over Peninsular India around 14 degrees N latitude; guides tropical cyclonic depressions from the Bay of Bengal into the mainland, determining spatial rainfall distribution. **[1.5 Marks]**

4. Why does the Coromandel Coast in Tamil Nadu receive winter rainfall while remaining relatively dry during the southwest monsoon? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Dry in Southwest Monsoon:** The Coromandel Coast lies parallel to the Bay of Bengal branch of the monsoon (preventing orographic lift) and is located in the leeward rain shadow of the Western Ghats for the Arabian Sea branch. **[1.5 Marks]**

• **Wet in Winter (Retreating Monsoon):** During October to December, the northeast retreating monsoon winds blow across the Bay of Bengal, picking up abundant moisture and yielding heavy rainfall upon striking the Tamil Nadu coast. **[1.5 Marks]**

5. Discuss the role of the Inter-Tropical Convergence Zone (ITCZ) and differential heating in generating the Southwest Monsoon of India. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Differential Heating:** In summer, the vast landmass of the Indian subcontinent heats up intensely compared to the Indian Ocean, establishing a deep thermal low-pressure center over north-western India. **[1 Mark]**

• **Shift of ITCZ:** The Inter-Tropical Convergence Zone (monsoon trough) shifts northward from the equator to approximately 20 to 25 degrees N latitude over the Indo-Gangetic Plain. **[1 Mark]**

• **Cross-Equatorial Flow:** The intense low pressure attracts the Southeast Trade Winds from the southern hemisphere across the equator; upon crossing the equator, they are deflected to their right by Coriolis force into the Southwest Monsoon winds. **[1 Mark]**

• **Bifurcation:** As these winds hit the Indian landmass, they bifurcate into two rain-bearing branches: the Arabian Sea branch and the Bay of Bengal branch. **[1 Mark]**

6. What is ENSO (El Nino-Southern Oscillation)? Explain how an El Nino event affects the Indian monsoon.
[Score: 4 Marks]

Answer & Key Valuation Steps:

• **ENSO Concept:** A coupled oceanic-atmospheric phenomenon involving periodic warming of sea surface temperatures in the central and eastern equatorial Pacific (El Nino) combined with seesaw pressure shifts between the Pacific and Indian Oceans (Southern Oscillation). **[2 Marks]**

• **Mechanism of Impact on Indian Monsoon:** El Nino shifts the rising branch of the Walker circulation eastward towards the central Pacific, creating a descending dry air branch over the Indian Ocean. **[1 Mark]**

• **Drought and Deficit Rainfall:** This atmospheric subsidence weakens the monsoon trough and delays the onset of monsoon winds, resulting in widespread drought and agricultural deficits across India. **[1 Mark]**